

Assignment 2- Case Study (8995-G level Task)

Stage 5 Tutorial Activities

Architecture and Responsibility

Week 8 | 60 minutes

Activity 1 - Where Does This Belong?

Responsibility	Layer	Reason
Read menu input	Infrastructure/Data	Reads internal storage system
Check appointment status transition	Domain	A technical component that does not operate concerning storage systems.
Coordinate booking use case	Application	Decides upon the order and arrangement of various inputs and their translation to main code logic.
Execute SQLite INSERT	Infrastructure/Data	Has a relation to SQL databases.
Format confirmation message	Infrastructure/Data	Infrastructure/Data is known for having access to external tools/APIs, meaning it could access email software.
Find appointment by ID	Domain	Identifying information and using it rather than storing it like infrastructure/data.

Activity 2 - Architecture Smell Hunt

A SmartCare file contains input(), SQL, appointment conflict rules, printing and validation. Identify at least five architecture problems and propose a layer for each responsibility.

The five architectural problems involved could include:

1. Translating data types for input () from the user
This problem can be classified in the responsibility layer for a software 'domain'.
2. Any syntax errors and logical errors perhaps involved with SQL
This problem can be classified in the responsibility layer for a software 'domain'.
3. Information is properly stored and collected in a third-party database
This problem can be classified in the responsibility layer for a software 'Infrastructure/data'.
4. Appointment booking session SQL table information is checked and validated
This problem can be classified in the responsibility layer for a software 'domain'.
5. Ensuring that if try/except is used, the specific syntax error is caught to prevent problems.
This problem can be classified in the responsibility layer for a software 'domain'.

Activity 3 - SOLID Without Overengineering

ClinicManager handles every use case. Which principle is threatened?

Single Responsibility Principle is challenged in the SOLID principles framework, as the rule clearly states that each class should have only one reason to change, meaning that it should not have multiple tasks/roles. This contradicts the possibility of there being too many connections from one actor.

AppointmentService imports sqlite3 directly. What dependency concern exists?

Liskov Substitution Principle states the need to be careful about object replacement for any inputs in a program.

A repository interface has 20 methods, but a client needs two. What concern exists?

Interface segregation principle is challenged, as it specifically states that there are some methods that do not need to be integrated in order to have many better interfaces rather than a bigger one.

Should every class have an interface? Explain.

Yes, if possible. The principle being challenged here is the Single Responsibility Principle, showing that each class should, in general, have at least one responsibility in software.

Activity 4 - AI Architecture Critique

AI proposes microservices, an event bus, six interfaces and a dependency-injection framework. Decide what to reject, defer or keep using the current requirements.

The suggested design includes microservices, an event bus, many interfaces, and dependency injection. Right now, the SmartCare system is small, and all its parts work closely together, so microservices and an event bus would make things harder without giving real benefits. Creating six interfaces is also unnecessary at this stage and can be added later if the system grows. A simple form of dependency injection is worth keeping because it helps organize the code and makes testing easier. This approach keeps the system straightforward while still allowing room to expand in the future